# LDL and 24 hours after Cardiac Arrest.

## Introduction

Low-density lipoprotein cholesterol (LDL-C) is a well-established risk factor for cardiovascular diseases, particularly atherosclerosis and coronary artery disease. However, emerging evidence suggests that LDL-C levels may also play a significant role in the outcomes of patients following cardiac arrest.

Understanding the dynamics of LDL-C within the first 24 hours post-cardiac arrest is crucial, as this period is critical for patient prognosis and management.

This literature review aims to synthesize current research on LDL-C levels observed within 24 hours after cardiac arrest, focusing on their association with patient outcomes.

## Methods

A comprehensive search was conducted using databases such as PubMed, MEDLINE, and Google Scholar to identify studies published up to February 2025.

Keywords used included "LDL cholesterol," "cardiac arrest," "post-cardiac arrest syndrome," and "lipid levels." Studies were selected based on their relevance, authenticity, and the availability of data on LDL-C levels within 24 hours post-cardiac arrest.

To investigate the relationship between low-density lipoprotein cholesterol (LDL-C) levels and patient outcomes within 24 hours post-cardiac arrest, a comprehensive literature search was conducted.

Inclusion criteria encompassed studies that specifically measured LDL-C levels within 24 hours following cardiac arrest and assessed associated patient outcomes. Both observational studies and clinical trials were considered.

Studies focusing solely on long-term lipid levels or those without specific data on the immediate 24-hour post-arrest period were excluded.

Data extraction focused on study design, patient demographics, LDL-C measurement timing, and reported outcomes. This methodology ensured a comprehensive analysis of current evidence regarding LDL-C levels in the immediate aftermath of cardiac arrest.

## Results

Several studies have reported significant changes in lipid profiles following acute cardiac events. For instance, a study by Pitt et al. (2008) observed that both total cholesterol and LDL-C levels decreased significantly in the 24 hours after admission for acute myocardial infarction (AMI), with reductions of 9% and 13%, respectively, by day four. High-density lipoprotein cholesterol (HDL-C) levels and triglycerides did not show significant changes during this period. Independent predictors of LDL-C decrease included the presence of diabetes mellitus and elevated cardiac troponin T levels. The authors concluded that LDL-C levels decrease significantly after an acute MI and that this reduction is correlated with cardiac troponin T levels (Pitt et al., 2008).

In the context of out-of-hospital cardiac arrest (OHCA), a retrospective observational study using the Korean Cardiac Arrest Resuscitation Consortium registry investigated the association between total serum cholesterol levels and outcomes upon discharge. The study found that patients with poor neurologic outcomes had significantly lower total serum cholesterol levels compared to those with good neurologic outcomes (127.5 ± 45.1 mg/dL vs. 155.1 ± 48.9 mg/dL, $p < 0.001$). Multivariate logistic regression analysis revealed that lower total serum cholesterol levels were associated with poor neurologic outcomes and in-hospital death, with odds ratios of 2.00 (95% CI 1.01–3.96, $p = 0.045$) and 1.72 (95% CI 1.15–2.57, $p = 0.009$), respectively (Ruggeri et al., 2022).

Another study examined the relationship between serum total cholesterol levels and neurologic outcomes in post-cardiac arrest patients. The findings suggested that lower cholesterol levels might indicate the degree of endotoxemia or inflammation caused by ischemic and reperfusion injury, potentially serving as a predictive marker for patient outcomes (Chae et al., 2022).

Conversely, some studies have highlighted that exceptionally low total cholesterol levels could be associated with an increased risk of OHCA. A case-control study demonstrated that individuals with very low total cholesterol levels had a higher risk of experiencing OHCA, suggesting that low cholesterol levels should be considered significant not only in estimating the prognosis of OHCA survivors but also in predicting the development of OHCA (Yang et al., 2021).

## Discussion

The observed decrease in LDL-C levels within 24 hours post-cardiac arrest may be attributed to several factors, including the acute phase response, hemodilution, and increased vascular permeability leading to lipid extravasation.

The association between lower LDL-C levels and poor neurologic outcomes could be reflective of the severity of the systemic inflammatory response and ischemia-reperfusion injury following cardiac arrest. However, the relationship between cholesterol levels and cardiac arrest outcomes is complex.

While high LDL-C is a known risk factor for coronary artery disease, which can lead to cardiac arrest, extremely low cholesterol levels have also been associated with adverse outcomes, possibly due to their role in cellular membrane integrity and hormone synthesis.

Therefore, both high and low extremes of cholesterol levels may be detrimental, and the optimal range for LDL-C in the context of cardiac arrest recovery remains to be determined.

## Conclusion

Current evidence indicates that LDL-C levels decrease within the first 24 hours following cardiac arrest, and lower levels are associated with poorer neurologic outcomes and increased in-hospital mortality.

Monitoring LDL-C levels in the immediate post-arrest period may provide valuable prognostic information and help guide therapeutic interventions.

Further research is needed to elucidate the mechanisms underlying these observations and to determine whether interventions aimed at modulating LDL-C levels can improve outcomes in post-cardiac arrest patients.